

Exploratory Data Analysis (EDA) Report – Titanic Dataset

Objective

The objective of this task was to perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn. The analysis aimed to understand the structure of the dataset, identify missing values, detect patterns and relationships, visualize important features, and summarize key insights.

Dataset Overview

The Titanic dataset contains information about passengers such as their age, gender, passenger class, fare, survival status, and other details. The dataset consists of 891 passenger records and 12 features.

Dataset Information

The dataset was examined using the `info()` and `describe()` functions. The analysis showed that some columns contain missing values. In particular, the **Cabin** column has a large number of missing entries, while the **Age** and **Embarked** columns also contain some missing values. The statistical summary provided information such as the mean, minimum, maximum, and standard deviation for numerical features.

Missing Values Analysis

The missing value analysis revealed that:

- The **Cabin** column contains the highest number of missing values and is incomplete for most passengers.
- The **Age** column has several missing values.
- The **Embarked** column has only a few missing values.
- Most other columns are complete and do not contain missing data.

Survival Distribution

The count plot of the **Survived** column showed that the number of passengers who did not survive was greater than the number of passengers who survived. This indicates that the majority of passengers lost their lives during the disaster.

Age Distribution

The histogram of the **Age** column showed that most passengers were between **20 and 40 years old**. There were relatively fewer children and elderly passengers. The distribution also indicated some missing age values.

Fare Distribution

The histogram and boxplot of the **Fare** column indicated that the fare distribution is highly right-skewed. Most passengers paid relatively low fares, while a small number of passengers paid significantly higher fares, resulting in several outliers.

Outlier Detection

Boxplots were used to identify outliers in numerical variables.

- The **Age** column contains a few outliers representing very young and very old passengers.
- The **Fare** column contains many extreme values because some passengers paid exceptionally high ticket prices.

Correlation Analysis

A correlation heatmap was created to examine relationships between numerical variables. The analysis showed:

- **Fare** has a positive correlation with passenger survival.
- **Passenger Class (Pclass)** has a negative correlation with survival, indicating that passengers in higher classes were more likely to survive.
- Other numerical variables showed weak or moderate correlations.

Pairplot Analysis

The pairplot helped visualize relationships among multiple numerical variables such as Age, Fare, Passenger Class, and Survival. It provided a clear overview of the distribution of each variable and the interaction between different variables.

Gender and Survival

The count plot comparing gender and survival revealed that **female passengers had a much higher survival rate than male passengers**. This suggests that women were given higher priority during the evacuation process.

Passenger Class and Survival

The analysis showed that passengers traveling in **First Class** had the highest survival rate, followed by those in Second Class. Third Class passengers experienced the lowest survival rate. This indicates that the passenger class had a significant influence on survival.

Scatter Plot Analysis

The scatter plot of Age versus Fare showed that passengers who paid higher fares generally had better survival chances. Although age alone did not strongly determine survival, fare and passenger class appeared to have a stronger relationship with survival.

Overall Findings

The exploratory data analysis provided several important insights:

- The dataset contains missing values, especially in the Cabin column.
- Most passengers were between 20 and 40 years of age.
- Ticket fares are highly skewed with several extreme values.
- Female passengers had a significantly higher survival rate than male passengers.
- First Class passengers were more likely to survive than passengers in lower classes.
- Higher ticket fares were associated with a greater probability of survival.
- Passenger Class and Fare were among the most influential features related to survival.
- Visualizations such as histograms, boxplots, scatter plots, pairplots, and heatmaps helped identify trends, relationships, and anomalies within the dataset.

Conclusion

The Exploratory Data Analysis successfully identified important characteristics of the Titanic dataset. The analysis highlighted the impact of gender, passenger class, and ticket fare on survival. Missing values and outliers were also identified, providing a better understanding of the dataset before applying any machine learning techniques. Overall, EDA proved to be an essential step in discovering meaningful insights and preparing the data for further analysis.